Windows Forensic Challenge Report

Subject: Static Analysis of **calc.exe**

Analyst: [Your Name]
Toolbox: DIE, Strings, PE-Bear

Date: May 14, 2026
Status: Completed

Phase 1: Identifying the Target

Initial assessment was performed using **Detect It Easy (DIE)** to determine the file's nature and complexity.

- **Entropy:** The entropy graph showed a low/stable value, indicating the binary is **not packed** or encrypted.
- **Entry Point:** Identified as the standard starting address for a Windows GUI application.
- **Compiler:** The binary was linked using **Microsoft Linker (14.xx)**, confirming it is a standard C++ Windows application.

[SCREENSHOT PLACEHOLDER]

Insert DIE Entropy Graph & Main Dashboard Screenshot here

Phase 2: String Extraction (Manual Analysis)

Using the `strings64.exe` tool, we extracted human-readable characters to reveal intent.

- **The Noise:** Thousands of strings were filtered to find specific indicators.
- **Dependencies:** Core DLLs identified: `SHELL32.dll`, `KERNEL32.dll`, `msvcrt.dll`.
- **Network:** Searched for "http". Only found `schemas.microsoft.com` (Standard Metadata). No malicious callbacks detected.

[SCREENSHOT PLACEHOLDER]

Insert image_b11ffe.png here (CMD results for DLLs and HTTP)

Phase 3: PE Header Surgery

Deep dive into the file structure using **PE-Bear**.

- **Sections:** Analyzed `.text` (code) and `.data` (globals). Raw vs. Virtual addresses align with standard PE formatting.

- **Imports:** `KERNEL32.dll` imports include functions like `GetProcessHeap` and `GetCurrentProcessId`.

[SCREENSHOT PLACEHOLDER]

Insert PE-Bear Import Directory / Section Headers Screenshot here

Phase 4: Evidence & Discovery

1. Timestamp:

The binary's "TimeDateStamp" in the File Header reveals its exact compilation date.

2. Dependency Analysis:

All loaded DLLs are legitimate Microsoft system files. No suspicious-sounding or third-party unknown DLLs were discovered.

3. The Discovery:

A PDB (Program Database) path was found in the strings: **C:\Users\...\calc.pdb**. This is a common "Easter Egg" or developer path that leaks information about the build environment.